

# Physics-Based State Reconstruction and Event-Source Diagnosability With Sparse PMU Measurements

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**ABSTRACT** Recovering unmeasured bus voltages and identifying a disturbance source are different inverse problems. Sparse phasor measurement units (PMUs) may determine a requested voltage function even when two sources remain indistinguishable. We formulate both tasks with a hybrid differential-algebraic model that separates physical interventions from measurement-integrity faults. Functional observability gives the minimum-variance linear covariance of an estimable voltage target. Source diagnosability is the weighted distance between candidate-response manifolds after state and corruption nuisance directions are fitted, with a local curvature bound for nonlinear separation. Three frozen contracts test the resulting claims on IEEE 39-bus models. A causal PowerDynamics estimator reaches 0.009634% mean hidden-voltage error, and measurement-only physical recentering removes at least 97.4% of the operating-point error gap. A 16-source load-intervention study obtains second-order local residual scaling and finds that event-visible information predicts the empirical 90%-detection amplitude with Spearman  $\rho = -0.794$ . Exact source Top-1 falls from 100% for a 0.4% load change to 13.0% at 0.01%, although the latter remains statistically detectable. In a separate ANDES bank, a supervised hierarchy localizes only 3 of 303 decisions after complete source holdout. The evidence links physical information geometry to state and source difficulty while exposing distinct limits for event visibility, source resolution, and learned transfer. A full multi-family joint estimator remains unvalidated.

**INDEX TERMS** dynamic state estimation, event detection, functional observability, limited observability, phasor measurement units, physics-informed inference, source localization.

## I. INTRODUCTION

**P**HASOR measurement units (PMUs) reveal electromechanical transients through synchronized voltage and current phasors, yet transmission networks instrument only a fraction of their buses and branches [1], [2]. A disturbance may begin outside the measured region, alter hidden voltages, and propagate to the available devices. Missing frames and corrupted channels create a second ambiguity because they can resemble physical transients. Diagnosis must therefore separate three questions: what is the network state, which event occurred, and where did each applicable physical or measurement event originate?

Dynamic estimators combine network models with synchrophasors [2], [3]. Sparse-PMU classifiers identify events and faulted assets [4], [5]. Other learning methods reconstruct hidden trajectories before localization or estimate state

and attack location from partial observations [6], [7]. These methods solve different inverse problems. The unresolved case considered here has an under-observed state, concurrent physical and measurement-integrity changes, source assets absent from labeled training, and sensing geometries that may render two sources indistinguishable.

The distinction is operational, not semantic. A generation change can be visible throughout the network while its source remains ambiguous. A dropped PMU identifies an integrity location directly but removes information needed to localize a simultaneous physical event. Likewise, a state estimate can have small point error while the intervention that produced it is not uniquely identifiable. A single multiclass label conceals these differences and produces accuracy values whose denominators do not match the claimed task.

A physical candidate is represented by  $h = (c, \ell, \tau, \gamma)$ ,

containing the event family, source, onset, and magnitude. Measurement availability and corruption remain separate variables. Each candidate generates a counterfactual DAE response, while candidate comparison fits admissible initial-state and integrity nuisance directions. The distance that remains after this fit measures source separation. A zero distance identifies a pair for which a uniformly correct source decision is impossible under the declared local model.

Three frozen contracts supply complementary evidence. The PowerDynamics state study reconstructs 31 hidden complex bus voltages from eight voltage/current PMUs and tests operating-point mismatch. A second PowerDynamics contract uses candidate-specific nonlinear load interventions to infer event presence, amplitude, and one of 16 source buses without a fitted source-label head. The ANDES study evaluates a physics-guided hierarchical ExtraTrees baseline on eight abnormal labels plus normal operation. The two simulators have different semantics, state coordinates, and data splits; their metrics are never pooled.

The contributions are:

- 1) a hybrid DAE inference model that maps eight event labels to composable physical-intervention and measurement-integrity variables;
- 2) a finite-horizon analysis that separates target-state estimability from event-source diagnosability, including a minimum-variance covariance, a sparse-integrity margin, a nonlinear remainder bound, and source-specific detectability information; and
- 3) frozen tests of hidden-voltage reconstruction, operating-point recentering, and Bayesian load-source inference, accompanied by a complete-source-holdout audit of the earlier supervised hierarchy.

The empirical boundary is sharp. The state estimator reaches 0.009634% mean hidden-voltage HVRE, while physical recentering closes 97.4%–98.3% of the operating-point error gap. In the load-intervention study, the physical likelihood resolves all sources at 0.4% magnitude but only 13.0% at 0.01%. The supervised event model detects and classifies known-source disturbances, yet complete source holdout leaves 3 correct decisions among 303. Physical candidate inference is therefore supported for one matched event family, not for the full eight-family joint problem.

## II. PRIOR WORK AND RESEARCH GAP

### A. STATE, ALGEBRAIC VARIABLE, AND UNKNOWN-INPUT ESTIMATION

Power-system dynamic state estimation combines machine, control, load, and network models with high-rate synchronized measurements. The task-force review in [2] separates dynamic, tracking, forecasting-aided, and static formulations and identifies model uncertainty and measurement availability as persistent constraints. Gaussian-sum and event-triggered filters address outliers or non-Gaussian measurement noise [8], [9]; a stale network or equilibrium center is a different model error. Nugroho *et al.* construct observers

for differential and algebraic DAE states with unknown inputs and input-sensor failures [10]. Katanic *et al.* estimate the same state types in a partially known nonlinear DAE and derive a topological estimability condition [3]. Qu *et al.* derive an event-triggered joint state/unknown-input estimator for multi-machine grids [11], while Nguyen *et al.* use distributed state-input estimation in active networks [12]. Joint state/input estimation is therefore established. Those unknown inputs, however, do not define a multi-family physical-source bank with a simultaneous integrity state.

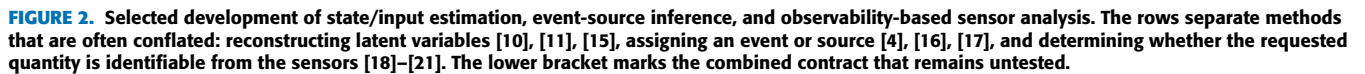
Learning methods recover states under sparse or heterogeneous measurements by encoding graph structure, power-flow relations, or a model-based estimator in the representation. Ngo *et al.* combine a graph neural network with physics-based state estimates [13]; Yue *et al.* couple graph learning to an outlier-resistant distribution-system estimator and include sparse-PMU line-outage identification [14]. Moshtagh *et al.* address PMU-unobservable transmission state estimation with a topology-aware graph architecture tested under topology change, PMU failure, bad data, and non-Gaussian noise [15]. These methods reconstruct state under incomplete or faulty sensing. Tolerating an outage does not, by itself, identify the outaged line.

### B. SPARSE-PMU EVENT AND SOURCE INFERENCE

PMU event identification has been addressed through subspace methods, engineered features, transfer learning, and graph models [22]–[25]. Cheng *et al.* demonstrate label-light online detection on a large field-PMU stream [26], while Abdel Hai *et al.* study transfer when only a small labeled set is available [27]. Their targets are event presence or family. Physical-source recovery is a separate output. Pandey *et al.* jointly detect, classify, and localize events using synchrophasor measurements and system information [4]. Sparse-PMU methods locate faulted lines, generation losses, or outages using learned models and electrical structure [5], [28]–[30]. Group-sparse dictionaries use admittance-, impedance-, and injection-based response models to localize several disturbance types with limited measurements [31]; a spatiotemporal graph model can also place event type, topology, and localization cost in its loss [17].

Open-set recognition differs from source holdout. Kummerow *et al.* use Siamese recurrent embeddings to reject event types or locations absent from training [32]; their subsequent sigmoid-network formulation learns class-dependent rejection boundaries under limited and full PMU coverage [16]. Hu and Cheng use generative sliding-window evidence and decision fusion to assign unseen events to a dedicated open category [33]. Open-set methods decide whether an observation falls outside a learned class set. Source generalization asks whether an untrained but physically enumerable bus or line can be evaluated as a counterfactual source. Rejection and physical candidate ranking answer different questions.

Benchmark availability does not remove the evaluation problem. The pmuBAGE collection offers almost one thousand privacy-preserving generated event records learned from



Limited PMU coverage is already a well-studied problem. The unresolved issue is the evaluation protocol. Many classifiers learn a fixed list of asset labels, so testing another replica at a known source measures interpolation rather than source generalization. Kummerow and Bretschneider show that measurement noise, unseen operating points, and unseen disturbances materially change the evaluation of learned PMU classifiers [37]. Zhai *et al.* reconstruct full state and localize line attacks from partial observations using an LSTM across multiple benchmark networks [7]. Rimkus *et al.* reconstruct missing trajectories without fault-location labels before applying a localization method [6]. Together these works establish reconstruction followed by localization as prior art. They do not jointly represent the eight physical/integrity states used here or test the held-source, diagnosability, and model-mismatch conditions together.

Functional observability asks whether a requested function of state can be inferred without full-state observability [19], [38]. Unknown-input observability instead asks whether exogenous inputs can be reconstructed from output histories [18]; sparse strong observability extends the question to corrupted sensor sets [39], [40]. PMU placement based on empirical observability Gramians similarly treats sensor value as task dependent [41], [42]. For event sources, Jamei *et al.* derive placement-dependent performance limits for fault localization [20]; Conte *et al.* prove sparse-PMU fault-localization conditions that partition a radial grid into indistinguishable line clusters [21]; and Yildiz and Abur optimize sparse placement for event detection and identification [43]. These results distinguish hidden-state reconstruction, input reconstruction, and pairwise source separation. They also rule out a general optimal-placement claim from one prescribed eight-PMU set.

Table 1 separates the capabilities needed for the combined problem. A source-agnostic physical hypothesis must be evaluated while the unobserved trajectory is estimated and measurement failures remain possible. The sensing geometry must expose indistinguishable sources before a classifier is trusted. Evaluation must then exclude the physical source

**TABLE 1. Capability comparison for the closest methodological families. “Held source” means that the physical asset is absent from source-labeled training or tuning. Entries describe published tasks and assumptions, not a cross-paper performance ranking.**

| Work                        | State target                            | Event/source target                                | Physics mechanism                                                      | Sparse/partial sensing     | Held source                  | Integrity separate                          |
|-----------------------------|-----------------------------------------|----------------------------------------------------|------------------------------------------------------------------------|----------------------------|------------------------------|---------------------------------------------|
| Katanic <i>et al.</i> [3]   | Differential and algebraic DAE states   | Unknown injectors, not event taxonomy              | Incomplete nonlinear DAE and least squares                             | Yes                        | N/A                          | No                                          |
| Nugroho <i>et al.</i> [10]  | Differential and algebraic DAE states   | Unknown input and input-sensor failure             | Outlier-resistant DAE observer                                         | PMU voltage/current        | N/A                          | Failure tolerance, not compositional source |
| Qu <i>et al.</i> [11]       | Machine state                           | Unknown field voltage and torque                   | Event-triggered nonlinear state/input estimator                        | Event-triggered PMUs       | N/A                          | Covariance bounds                           |
| Ngo <i>et al.</i> [13]      | Power-system states                     | None                                               | Branch-current physics plus graph learning                             | Yes                        | N/A                          | No                                          |
| Yue <i>et al.</i> [14]      | Distribution-system state               | Line outage                                        | Grid-model-informed graph learning and outlier-resistant SE            | Yes                        | Not stated                   | No                                          |
| Moshtagh <i>et al.</i> [15] | Transmission-system state               | Topology variation, not source output              | Topology-aware graph learning                                          | PMU-unobservable systems   | Not a source task            | PMU failure and bad-data tests              |
| Pandey <i>et al.</i> [4]    | None                                    | Detection, class, and source                       | Topology/equipment information                                         | PMU network                | Not stated                   | No                                          |
| Yildiz and Abur [5]         | None                                    | Fault detection and location                       | OLS/CNN with strategic PMU placement                                   | Yes                        | Not stated                   | No                                          |
| Li <i>et al.</i> [17]       | None                                    | Multi-family disturbance source                    | Topology/type-constrained STGCN loss                                   | Yes                        | Not stated                   | No                                          |
| Kummerow <i>et al.</i> [16] | None                                    | Known/unknown event type and location              | Siamese distance with learned rejection                                | Limited/full PMU sets      | Open-set rejection           | Rejection boundary                          |
| Hu and Cheng [33]           | None                                    | Open-set event detection/classification            | Generative window evidence and fusion                                  | Streaming PMUs             | Unknown-event category       | No physical-source posterior                |
| Rimkus <i>et al.</i> [6]    | Missing PMU trajectories                | Downstream fault localization                      | Graph reconstruction before diagnosis                                  | Yes                        | No source labels for imputer | Corruption via imputation                   |
| Zhai <i>et al.</i> [7]      | Full dynamical state                    | Line-attack location                               | Learned temporal dynamics                                              | Yes                        | Across benchmark contracts   | Attack target only                          |
| P07 method                  | Hidden voltage field; state uncertainty | Eight event labels; physical and integrity sources | DAE hypotheses, profiled separation, residual correction not evaluated | Eight prescribed PMU buses | Yes; learned baseline fails  | Yes                                         |

asset from all source-labeled fitting. Under this protocol, learning is useful only when it corrects model discrepancy or reduces candidate-search cost without memorizing asset labels.

The available evidence covers only part of this intersection. PowerDynamics supports hidden-voltage estimation and physical recentering. ANDES supports known-source event recognition and documents failure under source holdout. Joint candidate-conditioned inference and the predicted relation between separation and uncertainty still require a common experimental contract.

### III. JOINT STATE, EVENT, AND SOURCE PROBLEM

#### A. HYBRID DAE AND ELECTRICAL MEASUREMENT MODEL

Let  $x(t) \in \mathbb{R}^{n_x}$  denote differential states,  $z(t) \in \mathbb{R}^{n_z}$  algebraic states,  $u(t)$  scheduled inputs,  $q(t)$  network configuration, and  $\theta_k = \theta_0 + G_\alpha \alpha_k$  physical parameters with a slowly varying operating coordinate  $\alpha_k$ . A physical intervention  $a_h(t)$  is indexed by hypothesis

$$h = (c, \ell, \tau, \gamma, d), \quad h \in \mathcal{H}, \quad (1)$$

where  $c$  is event family,  $\ell$  is a bus or branch source,  $\tau$  is onset,  $\gamma$  is severity, and  $d$  describes the allowed temporal profile.

The plant is

$$\dot{x}(t) = f(x, z, u, q, a_h; \theta_k) + w_x(t), \quad (2)$$

$$\mathbf{0} = g(x, z, u, q, a_h; \theta_k), \quad (3)$$

$$y_k = M_k[h p_k(x_k, z_k, q_k; \theta_k) + b_k] + v_k. \quad (4)$$

Here  $M_k$  selects the channels that are available at time  $k$ ,  $b_k$  is a measurement-corruption process, and  $v_k$  is measurement noise. A dropout changes  $M_k$ ; bad data change  $b_k$ ; neither is a physical input to (2)–(3). This separation permits a physical event and a measurement failure to coexist.

The latent variables have deliberately different time scales and meanings. The pair  $(x, z)$  is the fast dynamic–algebraic state;  $\alpha$  is slow operating-point or parameter drift;  $a_h$  is an abrupt, sparse physical intervention;  $q$  is a discrete network mode; and  $(M, b)$  describes data availability and corruption. A colored model-discrepancy state  $c_k$  may enter either the process or measurement residual, as tested in Section V, but is not interchangeable with an event.

Within a smooth mode, assume the DAE is locally index one, so  $g_z$  is nonsingular after the reference-angle constraint is imposed. Differentiating the algebraic constraint gives

$$\dot{z} = -g_z^{-1} \left( g_x \dot{x} + g_u \dot{u} + g_\theta \dot{\theta} \right). \quad (5)$$

Equation (5) is not applied across a fault insertion or topology switch, where algebraic variables can jump and the model

must be segmented. Differential states remain continuous unless a simulator-specific reset map is declared.

The algebraic network core is

$$I = Y_{\text{bus}}(q, \theta)V, \quad S_i = P_i + jQ_i = V_i I_i^*. \quad (6)$$

Writing  $Y_{\text{bus}} = G + jB$  and  $V_i = |V_i|e^{j\vartheta_i}$  gives

$$P_i = |V_i| \sum_j |V_j| [G_{ij} \cos \vartheta_{ij} + B_{ij} \sin \vartheta_{ij}], \quad (7)$$

$$Q_i = |V_i| \sum_j |V_j| [G_{ij} \sin \vartheta_{ij} - B_{ij} \cos \vartheta_{ij}], \quad (8)$$

where  $\vartheta_{ij} = \vartheta_i - \vartheta_j$ . A line outage changes the branch primitives and therefore  $Y_{\text{bus}}$ ; a bus fault adds a shunt-admittance intervention; generation and load changes modify the appropriate active/reactive injection equations. For a PMU at one terminal of branch  $(i, j)$ ,

$$I_{ij} = Y_{ff,ij}V_i + Y_{ft,ij}V_j, \quad I_{ji} = Y_{tf,ij}V_i + Y_{tt,ij}V_j. \quad (9)$$

The orientation, transformer tap, line charging, and shunt convention in (9) are part of the measurement model; replacing terminal current with bus-injection current changes the likelihood.

At the acquisition boundary, native measurements can include  $V_{abc}$ ,  $I_{abc}$ , frequency, ROCOF, quality flags, and availability. Positive-sequence phasors define the balanced network likelihood used here. Negative- and zero-sequence components, when available, remain unbalance or integrity evidence rather than being mixed into the positive-sequence state. Ideal continuous-frequency channels satisfy

$$f_i = f_0 + \frac{1}{2\pi} \frac{d\angle V_i}{dt}, \quad \text{ROCOF}_i = \frac{1}{2\pi} \frac{d^2 \angle V_i}{dt^2}, \quad (10)$$

but an implemented likelihood must use the causal estimator and filtering semantics of the corresponding PMU backend.

## B. SPARSE PMUS AND REQUESTED STATE FUNCTION

Both evidence campaigns use PMU buses

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}, \quad |\mathcal{P}| = 8. \quad (11)$$

The unobserved set is  $\mathcal{U} = \{1, \dots, 39\} \setminus \mathcal{P}$ , with  $|\mathcal{U}| = 31$ . The state-estimation campaign stacks rectangular voltage and controlled terminal-current components,

$$\mathcal{Y}_{\mathcal{P},k} = \text{col}_{p \in \mathcal{P}} [\Re V_{p,k}, \Im V_{p,k}, \Re I_{p\ell,k}, \Im I_{p\ell,k}], \quad (12)$$

and requests the hidden complex-voltage function

$$\mathbf{z}_{\mathcal{U},k} = \text{col}_{i \in \mathcal{U}} [\Re V_{i,k}, \Im V_{i,k}] \in \mathbb{R}^{62}. \quad (13)$$

The event campaign also uses voltage angle, frequency, availability, and, in its longer-window implementation, current and ROCOF. Measurement semantics are kept campaign-specific rather than declaring channels equivalent across simulators.

## C. EIGHT EVENT LABELS AS COMPOSABLE STATES

Table 2 defines the eight abnormal labels in the historical event bank. The scientific variables are not the label numbers. Classes 1–4 are single physical interventions; classes 5 and 7 are measurement-integrity states; classes 6 and 8 are compositions. Class 8 is a prescribed mixed event in the available simulations, not evidence of open-set recognition.

Table 3 states how the four physical families enter the candidate DAE. It defines the operators without implying that the joint posterior has already solved every family.

The internal decision at time  $k$  is

$$(\hat{\mathbf{z}}_{\mathcal{U},k}, \Sigma_{\mathbf{z},k}, \hat{\mathbf{c}}_k, \hat{\ell}_k^{\text{phys}}, \hat{\mathbf{m}}_k, \hat{\ell}_k^{\text{meas}}, \mathcal{C}_k), \quad (14)$$

where  $\mathbf{m}_k$  denotes the integrity state and  $\mathcal{C}_k \subseteq \mathcal{H}$  is a credible candidate set. A location is empty when it does not apply. The candidate set is required because limited observability can make a forced Top-1 decision scientifically unjustified.

## D. CORRECTNESS AND EVALUATION UNITS

State accuracy uses the mean hidden-voltage relative complex error (HVRE)

$$\text{HVRE}^{(r)} = \frac{1}{T_r |\mathcal{U}|} \sum_{k=1}^{T_r} \sum_{i \in \mathcal{U}} \frac{|\hat{V}_{i,k} - V_{i,k}|}{\max(|V_{i,k}|, \epsilon)}. \quad (15)$$

This trajectory-and-bus aggregate is reported as a percentage. It is TVE-style but is not the standard per-frame synchrophasor TVE conformity statistic. Event detection, event-family macro-F1, exact physical Top-1/Top-3, and integrity-source Top-1 use their own declared populations. The joint correctness indicator is one only when every applicable component of (14) is correct; a correct event family cannot conceal a wrong source. Confidence intervals use an independently generated trajectory, event parent, or rebuilt plant as the statistical unit, never individual 30-Hz samples.

Causal inference permits only  $y_{0,k}$  at time  $k$ . A fixed-lag smoother, complete-window statistic, or centered temporal feature is retrospective and is labeled accordingly. Source generalization requires that physical source  $\ell$  be absent from all source-labeled training and tuning, not merely that the test trajectory use a new random seed.

## IV. RECOVERABILITY AND DIAGNOSABILITY UNDER SPARSE PMUS

### A. FUNCTIONAL OBSERVABILITY OF THE REQUESTED VOLTAGE FIELD

Linearizing the index-reduced DAE around a verified equilibrium and sampling at  $\Delta t = 1/30$  s gives

$$\delta x_{k+1} = A_d \delta x_k + w_k, \quad (16)$$

$$\delta y_k = C \delta x_k + v_k, \quad (17)$$

$$\delta \mathbf{z}_{\mathcal{U},k} = L_{\mathcal{U}} \delta x_k. \quad (18)$$

The reduction assumes a regular local algebraic Jacobian after the reference mode is fixed. Here  $\delta x_k \in \mathbb{R}^{114}$  contains reduced differential coordinates,  $C$  maps them to 32 rectangular voltage/current PMU channels, and  $L_{\mathcal{U}}$  maps them to 62

**TABLE 2.** Event ontology and historical ANDES inventory. Each source configuration has five replicas. “Physical source” and “integrity source” are separate outputs.

| ID    | Operational meaning                 | Physical intervention $c$    | Physical source $\ell$ | Integrity state/source        | Records |
|-------|-------------------------------------|------------------------------|------------------------|-------------------------------|---------|
| 0     | Normal operation                    | none                         | none                   | nominal                       | 5       |
| 1     | Bus fault                           | fault admittance             | bus                    | nominal                       | 195     |
| 2     | Line outage                         | topology removal             | branch                 | nominal                       | 170     |
| 3     | Generation change                   | active/reactive injection    | generator bus          | nominal                       | 50      |
| 4     | Load change                         | active/reactive withdrawal   | load bus               | nominal                       | 90      |
| 5     | Missing data                        | none                         | none                   | dropout / PMU                 | 40      |
| 6     | Generation change with missing data | generation change            | generator bus          | dropout / PMU                 | 50      |
| 7     | Bad data                            | none                         | none                   | corruption / PMU channel      | 40      |
| 8     | Prescribed mixed disturbance        | generation plus load changes | declared generator bus | measurement disturbance / PMU | 50      |
| Total |                                     |                              |                        |                               | 690     |

**TABLE 3.** Physical intervention operators in the candidate bank. Algebraic jumps are resolved by the post-event network constraints; differential-state resets are absent unless explicitly stated.

| Family            | DAE action at source $\ell$                                                                                        | Candidate support   | Severity coordinates                             | State continuity                 |
|-------------------|--------------------------------------------------------------------------------------------------------------------|---------------------|--------------------------------------------------|----------------------------------|
| Bus fault         | Add a shunt primitive, $Y_{bus}^+ = Y_{bus}^- + y_{fe} e_{\ell} e_{\ell}^T$ , for the declared clearing interval   | admissible buses    | complex fault admittance or resistance/reactance | $x$ continuous; $z$ may jump     |
| Line outage       | Set branch status $q_{ij} : 1 \mapsto 0$ and remove its tap, series, charging, and shunt primitives from $Y_{bus}$ | in-service branches | discrete status and switching time               | $x$ continuous; $z$ may jump     |
| Generation change | Modify scheduled active/reactive injection or a declared governor reference at generator bus $\ell$                | generator buses     | $\Delta P_G, \Delta Q_G$ , duration              | continuous input-driven response |
| Load change       | Modify the active/reactive withdrawal model at load bus $\ell$                                                     | load buses          | $\Delta P_L, \Delta Q_L$ , ZIP convention        | continuous input-driven response |

rectangular hidden-voltage components. These matrices are local and hypothesis dependent after a topology change.

For a horizon of  $N$  transitions, define

$$\mathcal{O}_N = \text{col}(C, CA_d, \dots, CA_d^N). \quad (19)$$

Full-state observability requires  $\text{rank}(\mathcal{O}_N) = 114$ . The requested voltage at time  $r \leq N$  instead has map  $L_r = L_{\mathcal{U}} A_d^r$ . Recovering the complete local target sequence uses  $\mathcal{L}_N = \text{col}(L_0, \dots, L_N)$ .

**Proposition 1 (Finite-horizon functional reconstruction):** For the noise-free local system, the target at time  $r$  is uniquely determined by the output sequence through  $N$  if and only if

$$\ker \mathcal{O}_N \subseteq \ker L_r. \quad (20)$$

The complete target sequence is uniquely determined if and only if  $\ker \mathcal{O}_N \subseteq \ker \mathcal{L}_N$ .

*Proof:* Two initial states produce the same output history exactly when their difference belongs to  $\ker \mathcal{O}_N$ . Their requested outputs at time  $r$  agree for every such pair exactly when the difference also belongs to  $\ker L_r$ . Stacking the maps gives the sequence statement. Equivalently, the rows of the requested map lie in the row space of  $\mathcal{O}_N$ . ■

This standard functional-observability result [19], [38] fixes the target and time index before any estimator is assessed. It also shows why a low full-state rank alone does not preclude recovery of the hidden-voltage function.

The noisy case gives a sharper statement than a rank test. Stack the output perturbations as  $\delta Y_N = \mathcal{O}_N \delta x_0 + v_N$ , assume  $v_N \sim \mathcal{N}(\mathbf{0}, R_N)$  with  $R_N \succ 0$ , and define  $G_N = \mathcal{O}_N^T R_N^{-1} \mathcal{O}_N$ .

**Corollary 1 (Minimum-variance functional estimate):** If (20) holds, the best linear unbiased estimate of the time- $r$  target and its error covariance are

$$\hat{\delta z}_{\mathcal{U},r} = L_r G_N^{\dagger} \mathcal{O}_N^T R_N^{-1} \delta Y_N, \quad (21)$$

$$\Sigma_{z,r}^* = L_r G_N^{\dagger} L_r^T. \quad (22)$$

Every other linear unbiased estimator has covariance  $\Sigma_{z,r}^* + QR_N Q^T$  for some  $Q$  satisfying  $Q\mathcal{O}_N = \mathbf{0}$ .

*Proof:* Condition (20) implies  $L_r = L_r G_N^{\dagger} G_N$ , which proves unbiasedness of (21). Direct substitution gives (22). Any other unbiased linear map is the displayed map plus  $Q$  with  $Q\mathcal{O}_N = \mathbf{0}$ . The covariance cross terms vanish because  $\mathcal{O}_N^T Q^T = \mathbf{0}$ , leaving the positive-semidefinite term  $QR_N Q^T$ . ■

For whitened measurements,  $\text{tr}(\Sigma_{z,r}^*)$  quantifies noise amplification in the requested voltage field. It depends on singular values and target alignment, not rank alone. The numerical curve below is an implementation of this quantity with a tolerance-dependent pseudoinverse; it is a conditioning diagnostic rather than an exact certificate.

## B. NUMERICAL AUDIT FOR THE PRESCRIBED EIGHT-PMU CASE

For the anchor-time target used by the structural audit, define

$$\Pi_N = \mathcal{O}_N^{\dagger} \mathcal{O}_N, \quad (23)$$

$$r_F(N) = \|L_{\mathcal{U}}(I - \Pi_N)\|_F, \quad (24)$$

$$r_{F,i}(N) = \|L_i(I - \Pi_N)\|_F. \quad (25)$$

A zero residual is an exact numerical condition only when the matrix scaling and rank decision are defensible. A nonzero value certifies that the computed observable subspace does not contain every requested anchor-time direction.

The implementation formed  $\mathcal{O}_N^\top \mathcal{O}_N$  and retained eigenvalues above  $10^{-9}$  of its maximum. Figure 3 turns the audit into a claim test. The pseudo-rank reaches 54 at  $N = 30$  but falls to 39 at  $N = 180$ , although the exact rank of a vertically augmented observability matrix cannot decrease. Gramian squaring, unstable-mode scaling, and a relative threshold therefore make this pseudo-rank unsuitable as an algebraic certificate. Longer history still lowers the trace-style noise bound by more than five orders of magnitude, but  $r_F(N)$  remains nonzero. The evidence supports improved numerical conditioning, not full-state or exact target observability. Empirical voltage recovery is evaluated separately and is not used to repair the failed certificate.

Horizons  $N \in \{0, 3, 10, 30, 60, 120, 180\}$  correspond to 0–6 s. The nonlinear nominal trajectories contain 91 frames, or 3 s, so  $N = 180$  is not a matched causal horizon for those records. Its bus ranking was nevertheless exported before nonlinear scoring and is retained only as a preregistered, cross-horizon structural diagnostic. The empirical test correlates  $r_{F,i}(180)$  with bus-level HVRE; it does not convert the six-second local audit into a three-second observability proof.

### C. EVENT-SOURCE DIAGNOSABILITY FROM CANDIDATE MANIFOLDS

State recoverability and source diagnosability have different nuisance spaces. Under candidate  $h$ , stack the available channels over  $H$  samples as

$$Y_H = s_{h,H} + \mathcal{O}_{h,H}\xi_h + \mathcal{E}_{S,H}\beta_h + \varepsilon_H, \quad \varepsilon_H \sim \mathcal{N}(\mathbf{0}, R_H), \quad (26)$$

where  $s_{h,H}$  is the intervention response,  $\xi_h$  is the locally admissible initial-state deviation, and  $\mathcal{E}_{S,H}\beta_h$  is corruption on a fixed support  $S$ . Allowing each candidate to refit nuisance variables produces the affine mean set

$$\mathcal{M}_h(S) = s_{h,H} + \text{col}[\mathcal{O}_{h,H}, \mathcal{E}_{S,H}]. \quad (27)$$

This candidate-specific form is required for topology-changing events because  $\mathcal{O}_{i,H}$  and  $\mathcal{O}_{j,H}$  need not agree.

Let  $\Delta s_{ij} = s_{i,H} - s_{j,H}$  and

$$\mathcal{N}_{ij,S} = [\mathcal{O}_{i,H}, -\mathcal{O}_{j,H}, \mathcal{E}_{S,H}]. \quad (28)$$

The  $R_H^{-1}$ -orthogonal nuisance projector and the pairwise separation margin are

$$P_{ij,S}^\perp = I - \mathcal{N}_{ij,S}(\mathcal{N}_{ij,S}^\top R_H^{-1} \mathcal{N}_{ij,S})^\dagger \mathcal{N}_{ij,S}^\top R_H^{-1}, \quad (29)$$

$$D_{ij,S}^{(\mathcal{P})}(H) = \frac{1}{2} \|P_{ij,S}^\perp \Delta s_{ij}\|_{R_H^{-1}}^2. \quad (30)$$

Equation (30) is half the minimum squared weighted distance between the two affine manifolds. It becomes the Gaussian mean-separation term only after nuisance values are fixed at the closest points. If all candidates are conditioned on one common pre-event state, the corresponding model replaces

the two independent state blocks in (28) by  $\mathcal{O}_{i,H} - \mathcal{O}_{j,H}$ . The analysis below uses independent candidate-specific state blocks; it does not use the common-state alternative.

**Proposition 2 (Local composite-hypothesis indistinguishability):** If  $D_{ij,S}^{(\mathcal{P})}(H) = 0$ , then  $\mathcal{M}_i(S)$  and  $\mathcal{M}_j(S)$  intersect. Hence there exist admissible nuisance values under candidates  $i$  and  $j$  that induce the same Gaussian measurement law. No decision rule can distinguish the pair uniformly over the declared nuisance set; with equal prior mass on those two parameter points, its average error is at least  $1/2$ .

*Proof:* The margin is zero exactly when  $\Delta s_{ij} \in \text{col}(\mathcal{N}_{ij,S})$ , which is equivalent to an intersection of the two affine mean sets. At the intersecting nuisance values, the means and covariance are identical. The observation law is therefore identical under both parameter points, so the two conditional error probabilities sum to one. ■

When the corruption support is unknown but at most  $q$  channels are affected, the conservative margin is

$$D_{ij}^{(q)}(H) = \min_{S: |S| \leq q} D_{ij,S}^{(\mathcal{P})}(H). \quad (31)$$

This union-of-subspaces formulation connects source localization to sparse strong observability and unknown-input reconstruction [18], [39], [40]. It also states the condition required for diagnosability under sparse corruption.

**Corollary 2 (Monotonicity under an added PMU):** For the same pair, covariance model, nuisance specification, and integrity budget, adding a conditionally independent PMU block cannot reduce the profiled margin:

$$D_{ij,S}^{(\mathcal{P} \cup \{p\})}(H) \geq D_{ij,S}^{(\mathcal{P})}(H). \quad (32)$$

For every nuisance vector, the augmented squared residual is the old nonnegative residual plus the new PMU residual. Taking the minimum preserves the inequality. Equality with a sum of separately profiled margins does not hold in general because the same nuisance variables must explain both sensor blocks; additivity requires fixed simple hypotheses or additional separability conditions.

The affine margin is useful only if it survives local DAE curvature. Let  $m_h(\eta_h)$  denote the exact, whitened, noise-free candidate response and suppose that, for  $\|\eta_h\|_2 \leq r$ ,

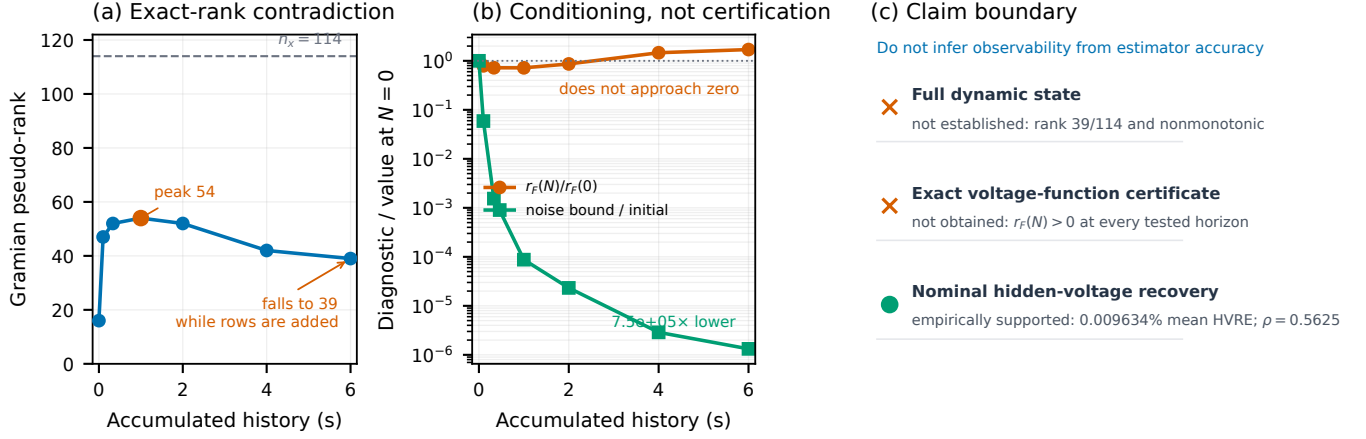
$$m_h(\eta_h) = s_h + N_h \eta_h + \rho_h(\eta_h), \quad \|\rho_h(\eta_h)\|_2 \leq \frac{\kappa_h}{2} \|\eta_h\|_2^2. \quad (33)$$

**Proposition 3 (Local nonlinear separation):** Let  $d_{ij}^{\text{lin}} = \sqrt{2D_{ij,S}^{(\mathcal{P})}(H)}$  and let  $d_{ij}^{\text{nl}}(r)$  be the minimum distance between the exact candidate responses over the two radius- $r$  nuisance balls. Under (33),

$$d_{ij}^{\text{nl}}(r) \geq \left[ d_{ij}^{\text{lin}} - \frac{\kappa_i + \kappa_j}{2} r^2 \right]_+. \quad (34)$$

Hence  $d_{ij}^{\text{lin}} > (\kappa_i + \kappa_j)r^2/2$  is sufficient for the two local nonlinear response sets to remain disjoint.

*Proof:* For any admissible pair  $(\eta_i, \eta_j)$ , the reverse triangle inequality bounds the exact distance by its affine part



**FIGURE 3.** What the prescribed eight-PMU audit permits one to claim. (a) The relative-threshold Gramian pseudo-rank decreases after  $N = 30$  even though samples are added, contradicting the monotonicity required of exact rank. (b) Values normalized to their  $N = 0$  levels separate two effects: accumulated history sharply lowers the trace-style noise bound, whereas the functional residual does not approach zero. (c) Consequently, the audit does not establish the full dynamic state or an exact hidden-voltage function. The small nominal TEST error is empirical evidence under the frozen experiment, not an observability proof.

minus  $\|\rho_i\|_2 + \|\rho_j\|_2$ . Minimizing the affine part and applying (33) gives (34); nonnegativity produces the positive-part operator.

This condition is sufficient, not necessary. A complete certificate requires directional curvature bounds for every candidate and nuisance direction. Section V-B tests the narrower load-intervention remainder and its second-order scaling. Curvature bounds for the full state, integrity, and multi-family candidate set remain unavailable.

For fixed closest nuisance values and unit-severity signature, the equal-prior Gaussian error is

$$P_{e,ij}^* = \Phi\left(-\sqrt{D_{ij}/2}\right), \quad |\gamma|_{\min} = \frac{\sqrt{2}\Phi^{-1}(1-\beta)}{\sqrt{D_{ij}(1)}} \quad (35)$$

to target  $P_e^* \leq \beta$  when the signature scales linearly with severity  $\gamma$ . This local severity limit is not applied across topology switches or nonlinear fault saturation. For simultaneous events with projected unit-signature matrix  $\tilde{S}_A$ , their amplitudes are locally identifiable only if  $\text{rank}(\tilde{S}_A) = |A|$  after the same nuisance projection.

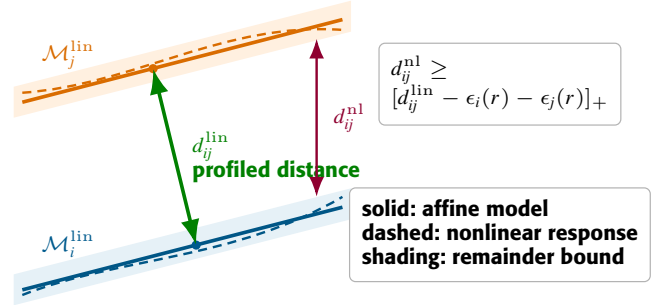
The executed load-intervention case gives a directly testable specialization. Let  $D_g$  be the whitened unit-amplitude response for candidate source  $g$  and let  $\Sigma$  be the normal-data covariance before whitening. Its event-visible information is

$$\mathcal{I}_g = D_g^T \Sigma^{-1} D_g, \quad |a|_{\min} = \frac{z_{1-\alpha} + z_{1-\beta}}{\sqrt{\mathcal{I}_g}}, \quad (36)$$

where the second expression is the local amplitude required by a one-sided Gaussian score with false-alarm probability  $\alpha$  and power  $1 - \beta$ . The formula predicts detectability, not source identity.

For source  $j$  as the closest single-signature alternative, the information left after its amplitude is fitted is

$$\mathcal{J}_{g \rightarrow j} = \mathcal{I}_g - \frac{(D_g^T \Sigma^{-1} D_j)^2}{D_j^T \Sigma^{-1} D_j} = \mathcal{I}_g(1 - \mu_{gj}^2), \quad (37)$$



**FIGURE 4.** Local source-diagnosability geometry. State and integrity nuisance generate one affine response set per candidate. The shortest weighted separation determines the linear margin. The shaded tubes represent bounded nonlinear remainders; overlap can erase a positive linear margin.

with  $\mu_{gj}$  the covariance-weighted signature coherence. Equation (37) is a pairwise local quantity. It need not predict multicandidate confusion once amplitude priors, unequal alternatives, and finite nonlinear response are present; the experiment retains that comparison as a falsifiable test.

For a calibrated source posterior, define the smallest candidate set satisfying

$$\mathcal{C}_k(\alpha) = \arg \min_{\mathcal{C} \subseteq \mathcal{H}} \left\{ |\mathcal{C}| : \sum_{h \in \mathcal{C}} p(h | y_{0:k}) \geq 1 - \alpha \right\}. \quad (38)$$

A diagnosability-aware system abstains from exact Top-1 when  $|\mathcal{C}_k| > 1$  or when its closest competing margin falls below a development-set threshold. The load-intervention study tests whether  $\mathcal{I}_g$  predicts detection amplitude and whether  $\mathcal{J}_{g \rightarrow j}$  predicts pair confusion. Candidate-set calibration, source-resolution delay, and PMU loss or addition still require a source-disjoint multi-family experiment.

## V. PHYSICS-BASED STATE ESTIMATION AND JOINT INFERENCE

### A. EXACT TARGET AND A GENERATIVE CANDIDATE MODEL

Let  $\zeta_k = L(x_k, z_k)$  be the electrical quantity requested by the application. Under squared loss, the exact Bayesian target is

$$\hat{\zeta}_k^{MMSE} = \mathbb{E}[\zeta_k | y_{0:k}], \quad \hat{h}_k^{MAP} = \arg \max_{h \in \mathcal{H}} p(h | y_{0:k}). \quad (39)$$

This statement is an optimality definition, not an implemented-algorithm claim.

For candidate  $h = (c, \ell, \tau, \gamma, d)$ , assume  $x_0 \sim \mathcal{N}(\bar{x}_0, P_0)$ , Gaussian process and measurement noise, the hard constraint  $g_h(x_t, z_t) = \mathbf{0}$ , and a Laplace prior on active corruption coefficients. With

$$r_{h,t} = M_t[y_t - h_{P_t}(x_t, z_t) - b_t], \quad (40)$$

the negative twice-log joint density conditional on  $h$ , up to constants and the declared corruption-support prior, is

$$\begin{aligned} \mathcal{J}_h^\circ = & \|x_0 - \bar{x}_0\|_{P_0^{-1}}^2 + \sum_{t=0}^k \|r_{h,t}\|_{R_t^{-1}}^2 \\ & + \sum_{t=0}^{k-1} \|x_{t+1} - F_h(x_t, z_t)\|_{Q^{-1}}^2 + 2\lambda_b \sum_{t=0}^k \|b_t\|_1, \end{aligned} \quad (41)$$

subject to the DAE constraints. Crucially,  $\mathcal{J}_h^\circ$  does not contain the hypothesis prior. The physical family and source enter through  $F_h, g_h$ , and the network/measurement operators rather than as an asset-label head.

The evidence is the constrained integral

$$p(y_{0:k} | h) = \int_{g_h=\mathbf{0}} \exp[-\mathcal{J}_h^\circ/2] dX dZ dB. \quad (42)$$

A Laplace approximation on locally identifiable directions gives

$$p(h | y_{0:k}) \propto p(h) \det^+(H_h)^{-1/2} \exp[-\mathcal{J}_h^\circ(\hat{X}_h, \hat{Z}_h, \hat{B}_h)/2], \quad (43)$$

so the prior appears exactly once. The pseudo-determinant requires the active constraint and corruption support to be fixed locally. If a Huber loss, a soft DAE penalty, or an omitted determinant is used without a corresponding normalized density, the result is a penalized evidence score, not a posterior probability.

The learned component is deliberately restricted. The earlier engineered feature system can propose a candidate set or learn a correction  $\psi_\phi(r_{h,0:k})$  to physical log evidence. It must consume candidate residuals, not source identity alone, and must be tested on held-out physical sources. The required ablation is learned-only, physics-only, physics plus learned discrepancy, and the complete diagnosability-aware rule.

### B. EVALUATED BAYESIAN LOAD-SOURCE MODEL

One restricted instance of the candidate model is executed rather than proposed. For each of 16 load buses, a centered finite difference of the nonlinear time-domain simulator forms a 30-frame, 32-channel response dictionary,

$$D_g = \frac{y_g(+\epsilon) - y_g(-\epsilon)}{2\epsilon}, \quad \epsilon = 0.005. \quad (44)$$

The dictionary is physical: it is regenerated by changing the load at candidate bus  $g$  and simulating the network. No source classifier or source label is fitted. Its frozen SHA-256 identifier is recorded in the repository manifest.

After subtracting the normal mean, the first-order event model is

$$r = aD_g + \epsilon, \quad a \sim \mathcal{N}(0, \sigma_a^2), \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \Sigma). \quad (45)$$

The covariance is estimated from normal CAL records. W0 uses identity covariance, W1 estimates channel covariance, and W2 adds a separable first-order temporal correlation. Under (45), the conditional amplitude posterior is Gaussian,

$$s_g^2 = (D_g^\top \Sigma^{-1} D_g + \sigma_a^{-2})^{-1}, \quad (46)$$

$$m_g = s_g^2 D_g^\top \Sigma^{-1} r. \quad (47)$$

The source evidence integrates  $a$  rather than inserting the true amplitude:

$$p(r | g) = \int p(r | a, g) p(a) da, \quad p(r | H_0) = \mathcal{N}(r; \mathbf{0}, \Sigma). \quad (48)$$

Normal versus event probability and the source posterior follow from these normalized densities and declared priors.

Finite changes bend the response. Symmetric development simulations estimate a source-specific second-order term  $Q_g$  and replace the event mean by

$$r_g(a) = aD_g + a^2 Q_g + O(a^3). \quad (49)$$

Four likelihood variants isolate mean curvature and amplitude-dependent covariance. DEV selects L2, which uses the second-order mean and the fixed W2 covariance; TEST then remains frozen. The analytic descriptor-DAE tangent is still incomplete, so (44) is described as a numerical physical tangent, not an analytic sensitivity.

### C. NOMINAL ESTIMATOR HIERARCHY

The estimator sequence changes one information source at a time. B0 retains only the equilibrium. B1 adds instantaneous PMU conditioning, B2 adds causal state propagation, and B3 tests retrospective temporal information. When the equilibrium becomes stale, E06-H estimates a physically admissible operating point through regularized nonlinear recentering. The full nonlinear Hybrid-DAE fixed-lag posterior is not evaluated in the experiments.

All nominal estimators use the same equilibrium  $y_0$ , hidden-voltage center  $z_{u,0}$ , sampled dynamics  $A_d$ , PMU map  $C$ , and hidden-output map  $L_u$ . This shared model isolates the value of measurement updating and temporal propagation.

The equilibrium baseline B0 returns

$$\hat{z}_{u,k}^{B0} = z_{u,0}. \quad (50)$$

It measures how much of the score can be obtained by remaining at the nominal power-flow solution.

The snapshot estimator B1 computes a prior-conditioned linear minimum-mean-square-error (LMMSE) update independently at every frame,

$$K_{\text{LMMSE}} = P_0 C^\top (C P_0 C^\top + R)^{-1}, \quad (51)$$

$$\hat{\delta x}_k^{\text{B1}} = K_{\text{LMMSE}}(y_k - y_0), \quad \hat{z}_{\mathcal{U},k}^{\text{B1}} = z_{\mathcal{U},0} + L_{\mathcal{U}} \hat{\delta x}_k^{\text{B1}}. \quad (52)$$

B1 uses network-constrained cross-covariance but no state carried from frame  $k - 1$ .

Conditional on one linear-Gaussian candidate with no corruption, B2 is not an ad hoc module: the Kalman recursion computes the exact conditional mean and covariance in (39). Candidate uncertainty, nonlinear switching, unknown integrity support, and non-Gaussian discrepancy are what make the enclosing hybrid problem approximate.

The causal Kalman estimator B2 applies the classical linear-Gaussian recursion [44]:

$$\hat{x}_{k|k-1} = A_d \hat{x}_{k-1|k-1}, \quad (53)$$

$$P_{k|k-1} = A_d P_{k-1|k-1} A_d^\top + Q, \quad (54)$$

$$\nu_k = (y_k - y_0) - C \hat{x}_{k|k-1}, \quad (55)$$

$$S_k = C P_{k|k-1} C^\top + R, \quad (56)$$

$$K_k = P_{k|k-1} C^\top S_k^{-1}, \quad (57)$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \nu_k. \quad (58)$$

The covariance update uses the Joseph form. The hidden estimate is  $z_{\mathcal{U},0} + L_{\mathcal{U}} \hat{x}_{k|k}$ . B2 is causal because only  $y_{0:k}$  enters the result at  $k$ .

B3 is an exact fixed-lag Gaussian smoother, following the linear Gaussian backward-recursion principle in [45], evaluated at lags  $\ell \in \{0, 1, 3, 5, 10, 15, 30, 60\}$  frames. It uses measurements through  $k + \ell$  to refine the estimate at  $k$  and is therefore retrospective by  $\ell/30$  seconds. Dense and cached implementations agree numerically on representative lags. B3 tests whether temporal information beyond the causal update improves the hidden-voltage target; it is not used to support a real-time claim.

#### D. MARGINAL CALIBRATION AND COLORED DISCREPANCY

For the hidden two-component voltage error at bus  $i$ , B2 provides

$$\Sigma_{i,k} = L_i P_{k|k} L_i^\top. \quad (59)$$

The original covariance is  $U_0$ . A two-block temperature model U2 is fitted on an independent calibration split:

$$\Sigma_{i,k}^{\text{U2}} = T \Sigma_{i,k} T^\top, \quad T = \text{diag}(\sqrt{\tau_{\mathcal{R}}}, \sqrt{\tau_{\mathcal{S}}}). \quad (60)$$

The temperatures minimize Gaussian negative log likelihood (NLL) on calibration data and are frozen before TEST. Coverage, NLL, normalized innovation squared (NIS), and temporal innovation correlation are reported separately; nominal marginal coverage alone is not called complete calibration.

The residual study considers a latent Gauss–Markov process

$$c_{k+1} = F_c c_k + \eta_k. \quad (61)$$

In D2-M,  $c_k$  enters the PMU measurement residual through a calibration-fitted low-rank basis. In D2-P, it enters a sparse physical process dictionary. Basis, rank,  $F_c$ , and latent covariance are selected using calibration records only. The experiment asks whether the observed low-rank structure improves reconstruction, hidden uncertainty, or innovation whiteness. It does not assume that a lower innovation autocorrelation is sufficient.

#### E. MODEL-ADEQUACY STATISTIC AND RECENTRING ABLATIONS

Let  $\mu_\nu$  and  $\Sigma_\nu$  be the mean and shrunk covariance of B2 innovations on the nominal calibration set. The per-frame adequacy statistic is the five-frame trailing mean of

$$a_k = (\nu_k - \mu_\nu)^\top \Sigma_\nu^{-1} (\nu_k - \mu_\nu). \quad (62)$$

Mismatch labels and hidden truth do not enter (62); hidden truth is used only after inference to label frames whose hidden HVRE exceeds the frozen evaluation threshold. AUROC and AUPRC therefore measure whether observed innovations anticipate unacceptable hidden error, not whether the detector identifies a mismatch family.

Three variants diagnose the failure mechanism without merging their evidence contracts. R1 replaces the nominal equilibrium with the true plant equilibrium and is explicitly oracle. R2-A estimates an eight-dimensional grouped observed-bus voltage offset using a 60-frame trailing window, nominal matrices, ridge regularization, and only PMU measurements. It is a causal negative ablation: its center/state cross-covariance is neglected and it does not enforce the nonlinear AC equations.

The corrected regularized physical estimator addresses M6 operating-point mismatch at one equilibrium snapshot. Let

$$d = \text{col}(\Delta P_{\text{PQ}}, \Delta Q_{\text{PQ}}, \Delta P_{\text{PV}}) \in \mathbb{R}^{43} \quad (63)$$

collect active and reactive changes at 17 admissible PQ loads and active-power changes at nine PV buses. Voltage is parameterized as  $V(\theta, \rho) = \exp(\rho + j\theta)$ . Defining the scaled PMU residual as  $e_y(V) = D_y^{-1}[h_{\mathcal{P}}(V) - y_{\mathcal{P}}]$ , the estimator solves

$$(\hat{V}, \hat{d}) = \arg \min_{V, d} \mathcal{J}(V, d), \quad (64)$$

$$\mathcal{J}(V, d) = \|e_y(V)\|_2^2 + \gamma^2 \|g_{\text{AC}}(V, d)\|_2^2 + \frac{\lambda}{q_d} \|d\|_2^2.$$

where  $h_{\mathcal{P}}$  is the audited voltage/terminal-current operator,  $g_{\text{AC}}$  includes PQ balance, explicit PV-magnitude constraints, solved PV reactive power, and bus 31 as Slack, and  $D_y$  scales voltage and current components separately. The weights are  $\gamma = 10^3$ ,  $q_d = 1$ , and  $\lambda = 0.01$ , selected on a disjoint DEV split; continuation initialization is then frozen for TEST. Neither the true nuisance vector nor the hidden voltages enter the TEST solve.

Because  $d$  may be non-unique, evaluation distinguishes nuisance information from target information. If  $J_{yd}$  and  $J_{zd}$  are the observed-PMU and hidden-voltage Jacobians with respect to  $d$  after eliminating voltage through the AC constraints, the local target residual is

$$r_{d \rightarrow z} = \frac{\|J_{zd}N(J_{yd})\|_F}{\max(\|J_{zd}\|_F, \epsilon)}, \quad (65)$$

where  $N(J_{yd})$  spans the PMU-uninformed nuisance nullspace. Exact target identifiability would require a zero residual. The experiment instead tests the weaker empirical question: whether (64) reconstructs the hidden voltage accurately despite rank-deficient nuisance information, and whether its Laplace covariance is credible.

## VI. VALIDATED BENCHMARK AND EXPERIMENTAL PROTOCOL

### A. BENCHMARK PROVENANCE AND NUMERICAL GATES

The primary plant uses Julia 1.11.9, PowerDynamics.jl 5.0.0, and NetworkDynamics.jl 1.3.0 [46]. It contains 39 buses, 46 branches, 19 loads, 10 sixth-order Sauer–Pai machines, nine AVR Type-I controllers, and nine TGOV1 governors. The descriptor realization has 192 coordinates: 114 differential and 78 algebraic. These parameters are package-defined tutorial data. We therefore classify the plant as a numerically validated pedagogical benchmark rather than an authoritative representation of an external grid.

The validation ladder checks the model before estimator error is scored. A same-plant static translation was audited after correcting tap orientation, shunts, line charging, and full- $Y_{bus}$  injection accounting. The largest remaining static discrepancy is  $1.71 \times 10^{-3}$  MVA in terminal flow. The controlled PMU operator agrees with the simulator to  $2.71 \times 10^{-7}$  pu in complex voltage and  $1.43 \times 10^{-14}$  pu in analytic/backend terminal current.

Dynamic validation checks the equilibrium, local modes, nonlinear trajectories, and linearization. The DAE equilibrium residual is  $4.66 \times 10^{-13}$ . After removing the reference mode, the least-damped physical pair is  $-0.09818 \pm j0.42948$  s $^{-1}$ . Nonlinear Rodas5P simulations remain finite for the audited perturbations. Amplitude sweeps satisfy

$$\|y_{NL}(\epsilon) - y_{LIN}(\epsilon)\|_2 \propto \epsilon^p, \quad (66)$$

with fitted slopes between 1.982 and 2.000 for representative state, governor-reference, and load perturbations, consistent with first-order linearization error.

### B. NOMINAL RECONSTRUCTION AND CALIBRATION

The nominal dataset contains 20 development and 100 test trajectories, each sampled at 30 Hz for 3 s. Perturbations are normal-operation changes to governor references, loads, or their combination. The equilibrium, linear model,  $P_0 = 10^{-2}I$ ,  $Q = 10^{-6}I$ , and  $R = 10^{-6}I$  are frozen before TEST. B0, B1, B2, and all B3 lags are evaluated on the same trajectories. A separate 30-trajectory calibration split

fits U1/U2 temperatures and low-rank discrepancy variants; TEST is not used for selection.

Hidden truth is used only for post-hoc metrics. The eight observed buses are excluded from every primary reconstruction score. The phasor prediction is formed as  $V_0 + \Delta V$ , and angle error is wrapped using  $\text{atan2}(\sin \Delta\theta, \cos \Delta\theta)$ . These rules correct an earlier exploratory implementation that incorrectly interpreted  $\angle \Delta V$  as the absolute angle; no superseded result from that implementation is retained.

### C. PHYSICAL MISMATCH CAMPAIGN

For each mismatch case, a fresh PowerDynamics plant is rebuilt from copied source tables, followed by power flow, dynamic initialization, and nonlinear simulation. The original package CSVs remain hash-identical. The seven families are network  $R/X/B$  (M1), machine parameters (M2), governor parameters (M3), AVR parameters (M4), ZIP/load model (M5), operating-point redispatch (M6), and coupled mismatch (M7). Stress scale is  $m \in \{0, 0.25, 0.5, 0.75, 1.0, 1.25, 1.5\}$ . These are engineering stress-test ranges, not probability priors.

All 980 main-grid cases and all 175 refinement cases passed power-flow, initialization, and time-domain acceptance. B1, B2, U0, U2, and the adequacy score are frozen from nominal data. Bootstrap intervals use seeds as independent units. The plots read the frozen per-case table directly; a previously used refined breakpoint mixed aggregation definitions and is excluded.

### D. ORACLE, OFFSET, AND PHYSICAL-MAP ADAPTATION CONTRACTS

The oracle subset compares the frozen estimator with true-equilibrium recentering and true local relinearization on the same mutated trajectories. Oracle quantities are evaluation-only. The offset-only recentering study uses new DEV seeds 101–110 and TEST seeds 201–220, disjoint from all earlier datasets. Its window, cadence, regularization, and activation threshold are selected on DEV and frozen before TEST. No event labels, hidden state, or mismatch family enter the R2-A estimate.

The corrected regularized physical-recentering campaign is a separate M6-only contract. It uses 30 fresh DEV cases from seeds 601–610 and 60 TEST cases from seeds 611–630, with 20 TEST equilibria at each of  $m \in \{0.5, 1.0, 1.5\}$ . DEV selects the nuisance scale, regularization, and continuation initialization from a small fixed grid. TEST uses only the observed voltage and terminal-current phasors; hidden voltages are revealed afterward for scoring. A solution is accepted only when the nonlinear optimizer succeeds, all outputs are finite, and the maximum AC residual is at most  $5 \times 10^{-5}$ . Rejections remain in the denominator; the frozen rejection table is empty. A six-case subset is also solved with an independent constrained optimizer, and its partial convergence status is reported rather than upgraded to full solver equivalence.

All numerical tables and empirical figures are regenerated from the hashed snapshot under `generated/frozen`. The

**TABLE 4. Frozen experimental contracts. State and event campaigns retain separate simulators, splits, targets, and denominators.**

| Study                            | Frozen contract                                                                                                 | Independent evaluation unit                           |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Structural audit                 | Preregistered voltage/current target residual, horizons 0–180                                                   | 31 hidden buses                                       |
| Nominal reconstruction           | 20 DEV / 100 TEST; 91 frames at 30 Hz; B0–B3 frozen                                                             | Complete nonlinear trajectory                         |
| Nominal calibration              | 30 independent CAL trajectories; U1/U2 and discrepancy only                                                     | Complete nonlinear trajectory                         |
| Physical mismatch                | Seven families, seven scales; B1/B2/U2 frozen                                                                   | 980 rebuilt plants                                    |
| Breakpoint refinement            | New seeds in transition bands; B2 frozen                                                                        | 175 rebuilt plants                                    |
| Causal recentering               | M1/M2/M6/M7; 160 DEV / 320 TEST; tune on DEV                                                                    | Rebuilt plant trajectory                              |
| Regularized physical recentering | M6 PQ/PV/Slack basis; 30 DEV / 60 TEST; one snapshot                                                            | Rebuilt equilibrium                                   |
| Load tangent audit               | 16 candidate load buses; centered $\pm 0.5\%$ nonlinear simulations                                             | Candidate-bus intervention                            |
| Bayesian load inference V1       | Eight amplitudes; 2,560 event / 320 no-event TEST records                                                       | Source–amplitude–noise record                         |
| Bayesian load inference V2       | 1,000 normal CAL; 500 normal + 640 event DEV; 9,600 weak + 800 no-event and 8,000 finite-amplitude TEST records | Source–amplitude–noise record                         |
| Known-source event diagnosis     | Eight abnormal labels plus normal; 414 TRAIN / 138 DEV / 138 TEST                                               | Complete ANDES trajectory                             |
| Source-transfer audit            | Asset-disjoint folds; true family supplied; exact source                                                        | 303 decisions from 101 event parents $\times$ 3 seeds |

build script checks the sample counts, acceptance flags, rank-deficient nuisance information, and preregistered adaptation-closure range before producing the tables and figures.

### E. LOAD-SOURCE INFERENCE CONTRACT

The load-source campaign uses the PowerDynamics plant and the same eight PMU buses, but it has its own frozen normal banks, event dictionary, and test manifests. The dictionary covers 16 candidate load buses listed in its manifest. Centered  $\pm 0.5\%$  nonlinear simulations define (44); the corresponding dictionary hash is fixed before the Bayesian tests. Candidate source and amplitude are inferred from the PMU window. The true source and true amplitude appear only in the evaluator.

V1 tests finite changes of  $\pm 1$ ,  $\pm 2.5$ ,  $\pm 5$ , and  $\pm 7.5\%$  on 2,560 event and 320 no-event records. V2 expands normal calibration to 1,000 records, uses 500 normal and 640 event records for likelihood selection, then opens two TEST blocks. The weak block contains 9,600 events at magnitudes from 0.01 to 0.4% and 800 no-event records. The finite block contains 8,000 events from  $\pm 1.25$  to  $\pm 9\%$ . Noise seeds are disjoint across CAL, DEV, and TEST. A test record is one source–amplitude–noise realization; repeated samples within its 30-frame window are never counted as independent cases.

W2 and L2 are selected on CAL/DEV, respectively, and frozen for both V2 TEST blocks. The weak-event operating threshold is 0.5 posterior event probability. No TEST row enters covariance estimation, likelihood selection, prior selection, or threshold tuning. The campaign covers one physical family and a fixed topology; it does not validate simultaneous events, missing PMUs, or cross-simulator transfer.

### F. EVENT-DIAGNOSIS CAMPAIGN AND SIGNAL EVIDENCE

The separate event campaign uses ANDES [47] and the same eight PMU buses. It contains 138 source configurations: one normal condition, 39 fault buses, 34 outage lines, ten generation connections, 18 load buses, eight missing-data PMUs, ten generation-and-missing pairs, eight bad-data PMUs, and ten prescribed mixed-event configurations. Five replicas per configuration give 690 complete trajectories. Each physical trajectory lasts approximately 5 s, with the event scheduled at 2 s; missing or corrupted intervals last 0.7 s. Physical in-

terventions are applied inside the simulator, whereas missing and corrupted values are applied to sampled measurements.

Three replicas per source configuration enter training, one development, and one test, producing 414/138/138 complete-trajectory splits. Consequently, the primary event benchmark evaluates new replicas at sources already represented in training. The separate asset-disjoint audit removes the physical source from all source-labeled fitting and tuning. Results from these contracts are reported side by side but never interpreted as the same generalization task.

The historical baseline is a hierarchy of extremely randomized trees. Its 136-feature version uses trailing voltage magnitude, angle, frequency, missingness, and a normalized disturbance score. The 448-feature version adds multi-scale statistics, adaptive residuals, and simultaneous contrasts among PMUs. A deterministic availability gate handles directly observed dropout. These are physics-guided features and typed decision heads; they are not a DAE-constrained learning architecture, and the ensemble’s leaf-vote scores are not treated as calibrated probabilities.

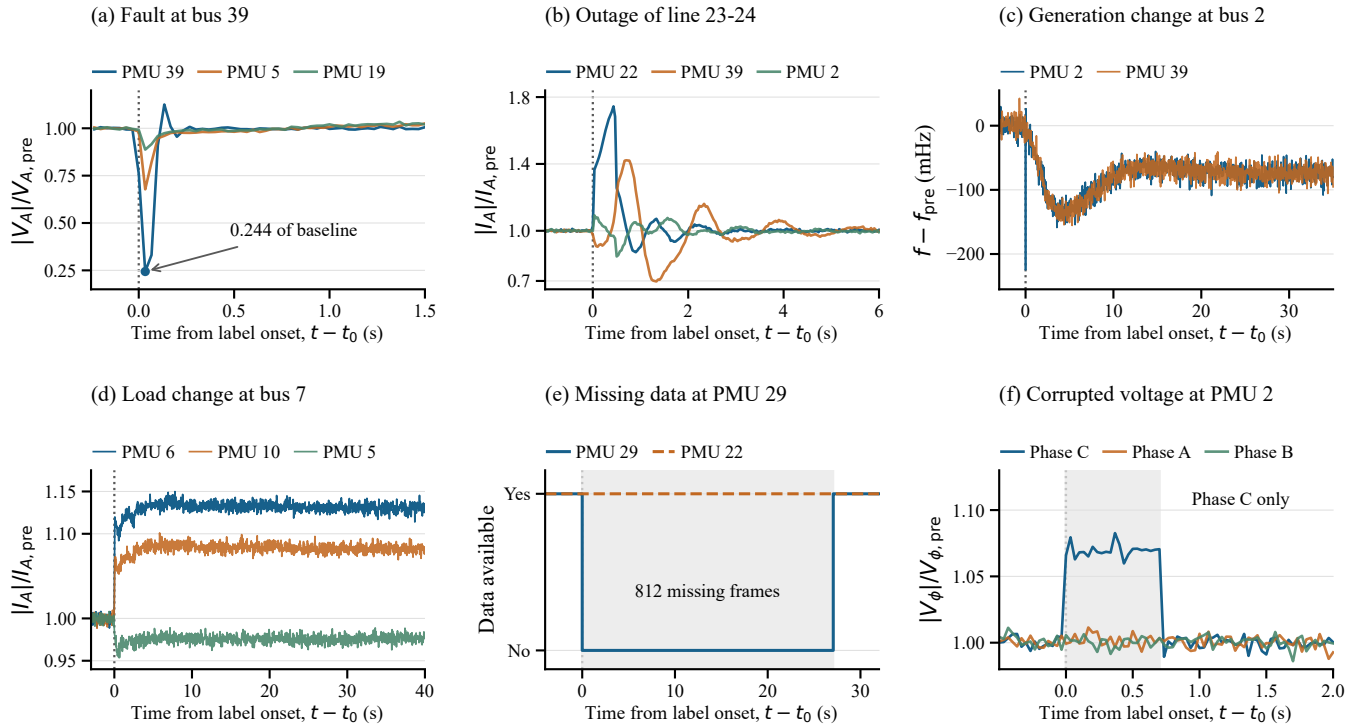
Figure 5 uses the first labeled episode of six representative families in the 89.65-min RAW0001 reference simulation. The curves are measured simulator outputs, not sketches or smoothed reconstructions. They show why no single channel suffices: a fault creates a short voltage sag, an outage redistributes current, injection changes have slower frequency/current responses, and measurement failures require availability or phase-consistency information.

## VII. NOMINAL RECONSTRUCTION AND STRUCTURAL EVIDENCE

### A. WHAT THE SPARSE PMUS RECOVER

Table 5 gives the nominal TEST results. Conditioning the equilibrium estimate on the current PMU sample (B1) reduces hidden-voltage HVRE by more than one order of magnitude. The causal Kalman recursion (B2) improves on B1 in 69.0% of the same trajectories, but its paired mean advantage is only  $-0.000270$  percentage points. Most of the nominal gain therefore comes from the measurement-conditioned physical map; temporal propagation supplies a smaller correction.

Figure 6(a) makes the trajectory distributions and scale separation visible, while panel (c) shows the paired differences



**FIGURE 5.** Representative measured event signatures from RAW0001. (a) Bus fault, (b) line outage, (c) generation change, (d) load change, (e) missing data, and (f) single-phase corruption. Dotted lines mark the first labeled sample; gray shading marks missing or corrupted intervals. Magnitudes are normalized by the preceding 3-s median, frequency is relative to its corresponding median, and time scales differ by panel. The plotted values and source-file hashes are frozen in the evidence snapshot.

**TABLE 5.** Nominal reconstruction on 100 frozen nonlinear TEST trajectories. HVRE is the trajectory-and-bus mean relative complex-voltage error over the 31 unobserved buses; intervals bootstrap independent trajectories.

| Method            | Complex RMSE | $ V $ RMSE | Angle RMSE ( $^\circ$ ) | Mean HVRE (%) | 95% HVRE interval (%) |
|-------------------|--------------|------------|-------------------------|---------------|-----------------------|
| B0 equilibrium    | 3.486e-03    | 2.496e-05  | 0.19497                 | 0.340200      | [0.293231, 0.387428]  |
| B1 snapshot LMMSE | 1.529e-04    | 5.091e-05  | 0.00821                 | 0.009904      | [0.008342, 0.011514]  |
| B2 causal Kalman  | 1.338e-04    | 4.235e-05  | 0.00722                 | 0.009634      | [0.008006, 0.011302]  |

rather than comparing overlapping marginal intervals. The equilibrium baseline is more than one order of magnitude worse than either measurement-updated estimator. This is not a comparison against learned or nonlinear DAE estimators; it is a controlled decomposition of equilibrium centering, instantaneous PMU conditioning, and causal temporal propagation within one local model.

### B. FUNCTIONAL STRUCTURE AS A PREDICTOR OF ERROR

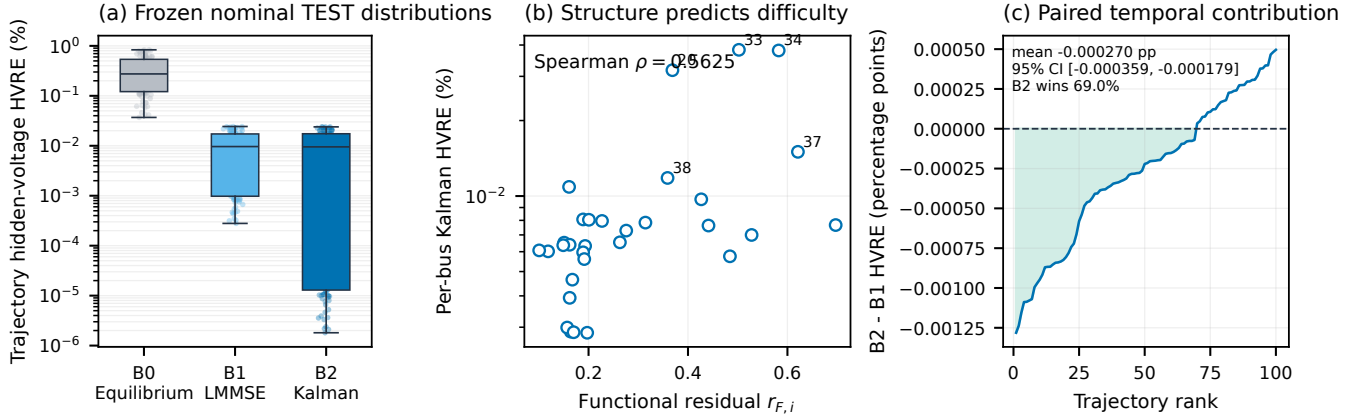
At horizon 180, the finite-horizon Gramian has pseudo-rank 39 of 114 and the aggregate target residual is nonzero; neither full-state nor exact functional observability is declared. The preregistered bus residual nevertheless correlates with B2 HVRE at Spearman  $\rho = 0.5625$  [Fig. 6(b)], with a two-sided permutation  $p$ -value of 0.0012. Because the 31 buses share trajectories and network coupling, bus-resampling intervals are treated as exploratory rather than as independent-unit confidence intervals. The same qualitative relation appears for B1.

The relationship is informative but incomplete. Bus 36 has

the largest structural residual, whereas the five largest B2 errors occur elsewhere. The projector captures one source of difficulty but not the nonlinear excitation distribution, estimator covariance, or every conditioning effect. Electrical distance to the nearest PMU has not been computed on the same audited network, so the analysis cannot yet show that  $r_{F,i}$  adds information beyond that confounder. The residual remains a diagnostic, not a binary observability certificate.

### C. RETROSPECTIVE SMOOTHING TEST

Across lags from one frame to 2 s, the exact fixed-lag smoother never improves mean HVRE over lag zero; every positive lag is slightly worse in the test results. Figure 7(a) shows the full curve. In this regime, current voltage and terminal-current phasors already dominate the hidden-voltage update, and additional future samples do not compensate for the local-model approximation. The negative result rules out latency as a useful trade for accuracy under the tested nominal perturbations.



**FIGURE 6. Nominal reconstruction and structural evidence.** (a) Trajectory-level hidden-voltage HVRE distributions on the same 100 nonlinear TEST trajectories. (b) Per-bus causal Kalman HVRE versus the horizon-180 functional residual exported before TEST scoring; labels identify the five buses with largest empirical error. (c) Sorted paired B2-minus-B1 HVRE differences; negative values favor the causal estimator, and the annotation reports the paired bootstrap interval and win rate.

## VIII. UNCERTAINTY, PHYSICAL MISMATCH, AND ADAPTATION

### A. MARGINAL CALIBRATION DOES NOT WHITEN INNOVATIONS

The nominal B2 covariance is conservative and covers essentially all real and imaginary errors at the nominal 95% level. Two temperatures fitted on 30 independent CAL trajectories bring the TEST marginal coverage closer to its target and improve hidden NLL (Table 6). They do not change the lag-one innovation autocorrelation, and the Ljung–Box test still rejects whiteness. Marginal calibration and temporal model adequacy are therefore separate properties [Fig. 7(b)–(c)].

Four principal components explain 99.2% of the calibration-residual variance, but low rank alone does not make the residual a useful process model. A rank-four measurement Gauss–Markov correction lowers the lag-one autocorrelation, then worsens HVRE and undercovers both voltage components. Placing one mode in the process model changes neither HVRE nor autocorrelation appreciably. Colored residual structure must therefore be tied to a physical discrepancy mechanism before it improves hidden-state uncertainty.

### B. MISMATCH RESPONSE DEPENDS ON MODEL FAMILY

The rebuilt-plant campaign reveals a clear family dependence (Fig. 8 and Table 7). At the largest tested scale, network mismatch raises the median error by  $16.99\times$  and an operating-point shift raises it by  $8.32\times$ . Machine, governor, AVR, and load-model perturbations remain near the nominal median. The estimator tolerates those tested component shifts but fails when the algebraic network map or equilibrium center changes.

Nominal covariance repair does not transfer automatically. Averaged across all mismatch cases, U0 has 92.9% 95% coverage and nominally fitted U2 has only 47.5%. The reduction reflects physical shift, not a calibration/test leak: U2 remains frozen from nominal CAL data. This result limits the

interpretation of Section V: nominal dispersion scaling is not robustness to a changed plant.

### C. ADEQUACY DETECTION AND PHYSICAL RECENTERING

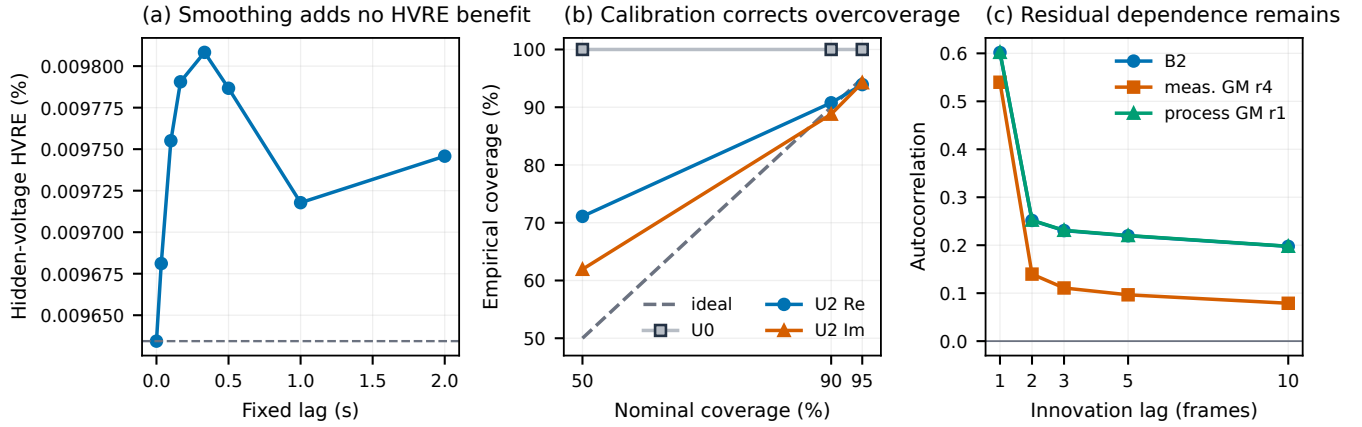
The observed-PMU innovation statistic detects many cases in which hidden HVRE is unacceptable. At the largest tested scale, its macro AUROC/AUPRC is 0.928/0.787 [Fig. 8(c)]. Family-level AUROC is omitted when every frame has the same frozen adequacy label; AUPRC remains defined. The statistic can flag a stale model, but it neither identifies the mismatch family nor supplies a calibrated alarm probability.

The earlier R2-A offset-only ablation remains scientifically useful because it isolates what does not solve the problem. Across nonnominal identifiable M1, M6, and M7 cases it closes only 0.4% of the oracle gap, despite passing its nominal-safety gate. That failure does not imply that adaptation from observed PMUs is impossible; it shows that grouped voltage offsets without AC balance and a rebuilt physical center are inadequate.

The regularized-recentering experiment tests that distinction for M6. Every TEST case satisfies the prespecified convergence gate. Across the three mismatch scales, the physical solve removes 97.4%–98.3% of the error gap to an oracle center [Fig. 9(a)–(b)]. The median solve takes 61.7 ms. Table 8 retains the point errors, paired intervals, and tail runtime; these static timings do not establish end-to-end real-time operation.

The nuisance information matrix has median rank 25 among 43 coordinates. Even with that nullspace, the requested voltage field is recovered with two orders of magnitude less error than the frozen center. This supports target recovery despite nuisance non-uniqueness. It does not prove exact functional identifiability because the local hidden-voltage residual remains nonzero, nor does it show that the injection vector is recovered.

Point accuracy does not certify posterior uncertainty. The nullspace-aware Laplace covariance covers every marginal



**FIGURE 7.** Temporal and uncertainty diagnostics on the nominal TEST set. (a) Exact fixed-lag smoothing; lag zero is the causal Kalman result. (b) Marginal real-component coverage for U0 and real/imaginary U2 blocks against nominal coverage. (c) Innovation autocorrelation for the original estimator and calibration-selected low-rank discrepancy placements.

**TABLE 6.** Frozen TEST uncertainty and colored-discrepancy ablations. U2 is fitted on a disjoint 30-trajectory calibration set; Gauss–Markov (GM) ranks are selected on that calibration set.

| Variant                | Hidden NLL | 95% cov. Re (%) | 95% cov. Im (%) | Innovation ACF(1) | HVRE (%) |
|------------------------|------------|-----------------|-----------------|-------------------|----------|
| U0 original            | -13.477    | 100.0           | 100.0           | 0.602             | 0.009634 |
| U2 Re/Im temperature   | -16.184    | 93.9            | 94.2            | 0.602             | 0.009634 |
| D2-M measurement GM r4 | 14.991     | 66.6            | 48.1            | 0.540             | 0.009788 |
| D2-P process GM r1     | -13.475    | 100.0           | 100.0           | 0.602             | 0.009638 |

**TABLE 7.** Median B2 hidden-voltage HVRE at the largest mismatch scale,  $m = 1.5$ . Ratios use the 140-case main-grid nominal median of 0.014744%.

| Mismatch family    | HVRE (%) | Ratio   |
|--------------------|----------|---------|
| M1 network         | 0.250430 | 16.985× |
| M2 machine         | 0.014744 | 1.000×  |
| M3 governor        | 0.014743 | 1.000×  |
| M4 AVR             | 0.014744 | 1.000×  |
| M5 load model      | 0.014761 | 1.001×  |
| M6 operating point | 0.122641 | 8.318×  |
| M7 coupled         | 0.023615 | 1.602×  |

interval, yet its joint NEES-like statistic grows with mismatch and remains far above the nominal reference [Fig. 9(c)]. The independent constrained solver also converges on only part of its small diagnostic subset. M6 point reconstruction is supported; joint uncertainty, network mismatch, coupled mismatch, and sequential adaptation are not.

#### D. PHYSICS-BASED LOAD DETECTION AND SOURCE RESOLUTION

The centered numerical tangent passes two checks before source inference is scored. Across all 16 load candidates, its largest discrepancy from either one-sided derivative is 0.235%, and the minimum cosine between the positive and negative perturbation directions is 0.999995. At a separate 10% stress, the median and maximum relative trajectory errors of the tangent prediction are 1.56% and 4.55%. These tests support the numerical physical dictionary over the stated range; they do not complete the pending analytic descriptor-

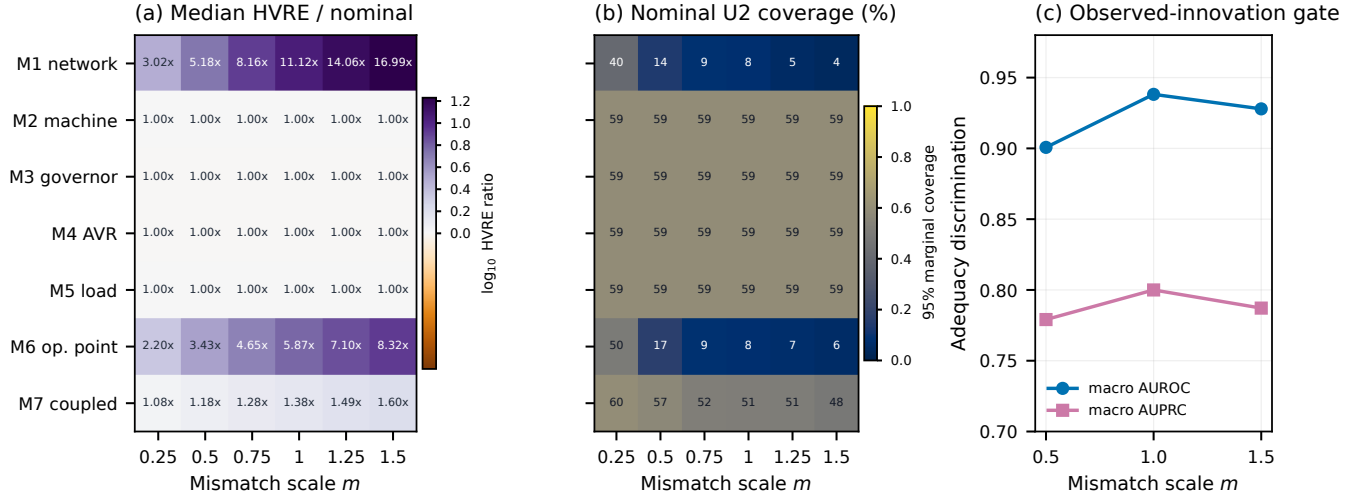
DAE derivative.

Normal-data covariance changes the decision rule materially. In V1, identity whitening produces a 100% false-positive rate and 10.9% source Top-1, whereas channel covariance plus temporal AR(1) whitening gives 0.625% false positives and 100.0% Top-1 on 2560 event records. The larger V2 normal bank confirms the statistical mechanism: W2 has NIS per frame 0.998, lag-one autocorrelation -0.0026, and Ljung–Box  $p = 0.598$ . Channel covariance alone leaves lag-one autocorrelation at 0.341.

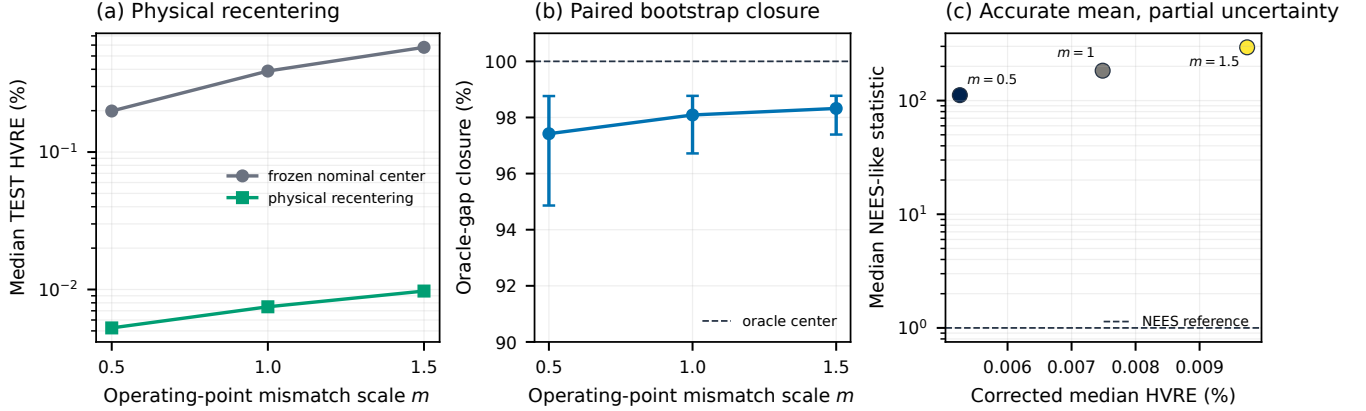
The nonlinear remainder follows the expected local order. A log–log fit over candidate buses and development amplitudes gives  $p = 2.00$  with bootstrap 95% interval [1.79, 2.24] [Fig. 10(a)]. The second-order mean model L2 is then selected on DEV with 95.0% amplitude coverage. On the independent finite-amplitude TEST block, its 95% coverage ranges from 92.9% to 96.0% over signed changes from 1.25 to 9% [Fig. 10(f)].

Weak events expose the difference between detection and localization. At  $|a| = 0.01\%$ , sign-pooled AUROC is 0.733 while exact source Top-1 is 13.0%; at  $|a| = 0.05\%$ , the same quantities are 0.995 and 73.4%. Source Top-1 reaches 100% at  $|a| = 0.4\%$  (Table 9). The event false-positive rate is 0.25% on 800 no-event records. Thus a window can contain detectable event evidence while remaining insufficient for a reliable source decision.

The theory predicts part, but not all, of that boundary. Source-wise event-visible information is negatively associated with the empirical amplitude required for 90% detection,



**FIGURE 8.** Physical-mismatch atlas for the fixed nominal estimator. (a) Median B2 hidden-voltage HVRE relative to the pooled  $m = 0$  median for every family-scale cell; annotations show the untransformed ratio and color encodes its base-10 logarithm. (b) Mean nominal-U2 marginal 95% coverage under the same rebuilt plants. (c) Macro adequacy discrimination from observed innovations; undefined family-level AUROCs are omitted from the macro mean. Each family-scale cell contains 20 independently rebuilt plants.



**FIGURE 9.** Corrected regularized physical recentering for held-out M6 operating-point mismatch. (a) Median hidden-voltage HVRE for the nominal center and the physical solve. (b) Paired oracle-gap closure with 95% bootstrap intervals over 20 independent TEST cases per scale; the oracle-center reference is 100%. (c) Corrected point error versus the median joint NEES-like diagnostic. Marginal 95% coverage is 100% at each scale, but the joint statistic remains far above its nominal reference.

with Spearman  $\rho = -0.794$  [Fig. 10(e)]. The projected pairwise information has only  $\rho = 0.361$  with empirical pair confusion and is retained as inconclusive. Event visibility is supported as a detectability predictor; the proposed pairwise source-confusion surrogate is not.

#### E. KNOWN-SOURCE DETECTION AND LOCALIZATION

On the known-source TEST split, the availability-gated hierarchy reaches detection F1 0.989 but only 0.592 physical Top-1 (Table 10). Expanding the representation adds 9.6 percentage points to source ranking over the compact hierarchy. Multiscale dynamics, adaptive residuals, and PMU contrasts therefore help when the source is represented during training; the availability rule contributes mainly to integrity handling.

The gap between detection F1 and physical Top-1 confirms that a recognizable disturbance need not have a uniquely recoverable source. False alarms decrease to 1.45 episodes

per normal-labeled minute, but the normal exposure is only about 7.15 min per split. That rate is evidence of alarm burden in a short synthetic stream, not an operational false-alarm guarantee.

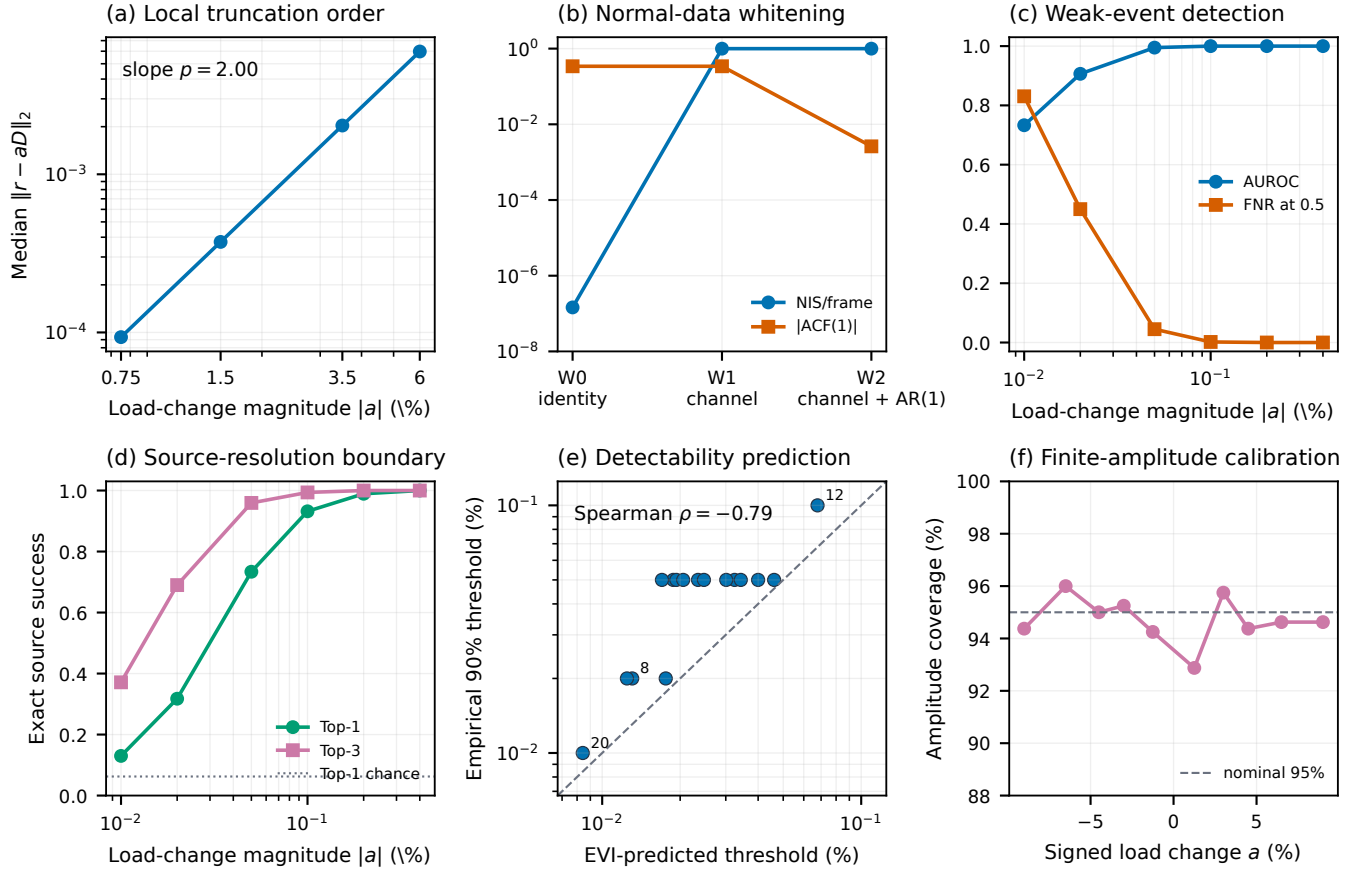
#### F. SOURCE HOLDOUT BREAKS LEARNED LOCALIZATION

With the physical source excluded from source-labeled fitting and the true event family supplied, only 3 of 303 decisions are exact. Detection and event-family confusion have been removed from this test. The remaining error shows that the learned location heads and candidate descriptors retain source-specific associations, invalidating a zero-shot-source claim for the legacy system.

Transfer to RAW0001 is also weak: the basic candidate model localizes only 0.3 of five physical episodes on average. The simulator bank omits some channels used by the longer-window implementation, so no single feature family can be

**TABLE 8.** Corrected physical static recentering on 60 held-out M6 operating-point cases. Gap-closure intervals bootstrap the 20 independent TEST cases at each scale. Rank and nullity refer to the 43-dimensional nuisance basis; runtime is per solve.

| $m$ | $n$ | Frozen HVRE (%) | Corrected HVRE (%) | Gap closure [95% CI] (%) | Rank  | Nullity | Median NEES | Runtime (ms) |
|-----|-----|-----------------|--------------------|--------------------------|-------|---------|-------------|--------------|
| 0.5 | 20  | 0.19885         | 0.00525            | 97.4 [94.9, 98.8]        | 25/43 | 18      | 111.6       | 57.6         |
| 1.0 | 20  | 0.38834         | 0.00749            | 98.1 [96.7, 98.8]        | 25/43 | 18      | 183.3       | 61.1         |
| 1.5 | 20  | 0.57795         | 0.00974            | 98.3 [97.4, 98.8]        | 25/43 | 18      | 294.2       | 69.2         |

**FIGURE 10.** Frozen physics-based load-event evidence. (a) First-order residual scaling over 16 candidate buses. (b) Normal-data covariance diagnostics for W0–W2. (c) Weak-event detection and (d) exact source resolution, pooled only across signs at the same magnitude. (e) Event-visible-information threshold against the empirical 90%-detection amplitude; labels identify representative source buses. (f) L2 amplitude coverage on the separate finite-amplitude TEST block. The weak block contains 9600 event and 800 no-event records; the finite block contains 8000 events.**TABLE 9.** Sign-pooled weak load-event TEST results for the frozen W2/L2 model. Each row contains 1,600 event cases (16 source buses, two signs, 50 noise replicas); the false-positive rate uses 800 no-event records.

| $ a $ (%) | AUROC | FPR (%) | FNR (%) | Top-1 (%) | Top-3 (%) |
|-----------|-------|---------|---------|-----------|-----------|
| 0.010     | 0.733 | 0.25    | 83.1    | 13.0      | 37.1      |
| 0.020     | 0.906 | 0.25    | 45.0    | 31.8      | 69.0      |
| 0.050     | 0.995 | 0.25    | 4.5     | 73.4      | 95.9      |
| 0.100     | 1.000 | 0.25    | 0.2     | 93.2      | 99.4      |
| 0.200     | 1.000 | 0.25    | 0.0     | 98.9      | 100.0     |
| 0.400     | 1.000 | 0.25    | 0.0     | 100.0     | 100.0     |

blamed for the deficit. Internal simulation accuracy does not establish cross-record reliability.

Supervised, physics-guided features work for detection and known-source diagnosis but not for source-generalizable in-

ference. The physical load experiment supplies a contrasting result: candidate sources are evaluated through simulated responses rather than learned asset labels. Its 16-source success is restricted to one event family, fixed topology, and matched simulator; it does not repair the broader held-source or cross-record failure.

## IX. DISCUSSION AND LIMITATIONS

The three contracts expose different parts of the inverse problem. Eight voltage/current PMUs recover the requested 31-bus voltage field accurately in the PowerDynamics state tests even though the 114-state local model is not fully observable. A separate physical load-response dictionary supports normalized event and source evidence on the same simulator. The ANDES bank measures what the earlier supervised

**TABLE 10.** Audited sequential event-diagnosis baseline on the known-source TEST split, mean over three fitted seeds. The independent split unit is the complete trajectory; timestamp-level scores retain the denominators of the frozen evaluator.

| Configuration               | Detection F1 | Event macro-F1 | Physical Top-1 | Physical Top-3 | Integrity Top-1 | False alarms/min |
|-----------------------------|--------------|----------------|----------------|----------------|-----------------|------------------|
| 136-feature hierarchy       | 0.965        | 0.791          | 0.496          | 0.646          | 0.870           | 4.01             |
| 448-feature hierarchy       | 0.975        | 0.815          | 0.590          | 0.717          | 0.865           | 1.45             |
| 448 features + availability | 0.989        | 0.905          | 0.592          | 0.719          | 0.974           | 1.45             |

representation retained when source labels were removed. Low hidden-voltage HVRE still does not guarantee source separation, and high event visibility does not guarantee exact localization.

The local theory explains this split. Proposition 1 removes unobservable directions that do not affect the requested voltage function. Corollary 1 then attaches a covariance to that estimable function; rank alone cannot provide it. Source diagnosis has a different geometry. Proposition 2 tests whether two candidate-nuisance response sets intersect, while Proposition 3 requires the affine distance to exceed the combined DAE remainder before local nonlinear separation is certified. Large faults, topology switches, and unmodeled controls may violate the neighborhood in which that bound holds.

Two theory-to-data links are now tested. The preregistered functional residual correlates with bus-level voltage error, although excitation, scaling, estimator covariance, and electrical distance explain several exceptions. In the load study, event-visible information predicts the source-wise detection threshold. The result has the expected negative sign: more whitened response energy requires a smaller amplitude for 90% detection. The projected pairwise information does not predict confusion convincingly. That negative test prevents a detectability result from being relabeled as a localization theory.

The operating-point study also separates target recovery from parameter recovery. Regularized physical recentering closes 98.1% of the oracle-center gap on M6 although only 25 of 43 nuisance directions are locally informed. The hidden-voltage target can therefore be accurate while the injection coordinates remain non-unique. This result is confined to an equilibrium snapshot. It does not cover line outages, faults, or sequential event-conditioned estimation, and the local curvature covariance remains under-dispersed in the joint NEES-like statistic.

The legacy event model supplies one reusable mechanism and one warning. Multiscale summaries, adaptive residuals, and simultaneous PMU contrasts improve known-source physical ranking by 9.6 percentage points. Availability belongs in the integrity state because a missing measurement is observed directly. Source identities and endpoint descriptors, however, allow memorization. Exact localization falls to 1.0% under complete physical-source holdout. The load experiment avoids an asset-label head by evaluating each candidate as a physical intervention. Learned residual features may still help, but their contribution must be measured against this physics-only likelihood.

The principal evidence gap is the absence of one multi-family end-to-end experiment. The state, load-source, and ANDES studies have different targets or simulator semantics. All are simulated; none supplies field or hardware-in-the-loop validation. The physical Bayesian result covers one intervention family, fixed topology, known onset window, and matched candidate simulator. The ANDES records also have fixed onsets and only about 7.15 min of normal-labeled exposure per split, too little for an operational false-alarm rate. PMU loss and addition have not been linked prospectively to the separation margin.

A decisive follow-up requires one audited plant and measurement operator for learned-only, physics-only, learned-residual, and diagnosability-aware methods across all physical families. Event parents and physical sources must be disjoint across fitting, calibration, and test. Operating-point shift, parameter error, topology change, and PMU loss or addition must be imposed before TEST is opened. A long normal stream must report false alarms per hour and event recall; event-source correctness, candidate-set coverage, resolution delay, HVRE, calibration error, runtime, and failed solves complete the evaluation. Until then, multi-family joint superiority remains unproved.

## X. CONCLUSION

Sparse-PMU monitoring contains two inverse problems with different information limits. Functional observability determines whether the requested hidden-voltage function is recoverable and gives the covariance of its minimum-variance linear estimate. Candidate-manifold separation determines whether two event sources remain distinguishable after state and integrity nuisance are fitted. A bounded DAE remainder connects that linear margin to a local nonlinear certificate.

The experiments show the distinction. The causal Power-Dynamics estimator attains 0.009634% mean hidden-voltage HVRE, and physical recentering removes 97.4%–98.3% of the operating-point error gap. In the executed 16-source load case, event-visible information predicts the 90%-detection amplitude, while source Top-1 falls from 100% at 0.4% magnitude to 13.0% at 0.01%. The separate ANDES model detects known-source events, yet complete physical-source holdout yields only 3 correct decisions among 303.

Physical candidate responses now have direct evidence for one matched event family. Extending them to faults, outages, generation changes, integrity events, changing PMU sets, and simulator shift remains the next source-disjoint test. The central consequence is narrower: state recoverability, event

detectability, and source diagnosability must be measured separately before any joint decision is trusted.

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